Smart Irrigation Rover: Project Overview

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Introduction to Robotics
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1. Research Methodology

1.1. Restatement of the Research Problem

The main problem addressed in this research is inefficient irrigation and limited automation in small-scale agricultural environments. In many traditional irrigation systems, water is applied manually or uniformly without properly checking the actual moisture condition of the soil. As a result, some areas may receive more water than necessary, while dry areas may remain untreated.

This project focuses on the idea of a Smart Irrigation Rover for Adaptive Spot Watering. The proposed system aims to use a mobile rover, soil moisture sensing, obstacle detection, and IoT-based monitoring to support more efficient irrigation. At this stage, the research mainly investigates the feasibility, design requirements, and user relevance of such a system through literature review and survey-based analysis.

1.2. Research Approach

This study follows a mixed-method research approach. The research combines secondary data from published literature with primary data collected through a Google Form survey.

The literature review was used to understand existing work in smart irrigation, agricultural robotics, IoT-based monitoring, soil moisture sensing, and autonomous navigation. The survey was used to understand user awareness, opinions, and practical expectations regarding smart irrigation and agricultural automation.

This approach was selected because the project is still in the planning and prototype development stage. Therefore, the methodology focuses on understanding the problem, identifying design requirements, and justifying the proposed solution before full hardware implementation and field testing.

1.3. Methodological Choice

The methodology is based on two major activities: literature review and survey analysis.

The literature review helped identify the technologies and methods commonly used in related systems. These include soil moisture sensors, IoT platforms, autonomous robots, water pump control, and decision-making methods for irrigation. The reviewed papers also helped identify research gaps such as high system cost, limited mobility, lack of real-time monitoring, and challenges in agricultural robot navigation.

The Google Form survey was used to collect practical feedback from respondents. This helped connect the research problem with real user needs. Since the project is intended to be a practical engineering solution, combining academic research with survey responses made the methodology more relevant and realistic.

1.4. Data Collection Process

Data for this research were collected from two sources.

First, secondary data were collected through a literature review of research papers related to smart agriculture, IoT-based irrigation, autonomous agricultural robots, soil moisture mapping, obstacle avoidance, and precision farming. The papers were selected based on their relevance to the project topic and their publication period, mainly focusing on recent works from 2022 to 2026.

Second, primary data were collected through a Google Form survey. The survey was prepared to understand general awareness of smart irrigation systems, problems in current irrigation practices, interest in automated irrigation, and opinions about using robotics and IoT in agriculture.

The survey responses were collected digitally and organized using Google Forms and spreadsheet tools. The collected responses will be used to prepare charts, percentages, and summary tables for the survey analysis section.

1.5. Data Analysis Method

The literature review data were analyzed using content-based analysis. The reviewed papers were grouped according to their main focus areas, such as smart irrigation, IoT monitoring, agricultural robotics, soil moisture sensing, AI-based irrigation decisions, and obstacle avoidance. From each paper, the main contribution, methodology, limitation, and future research direction were identified.

The survey data were analyzed using descriptive analysis. The responses from Google Forms were summarized using percentages, charts, and tables. This analysis helped identify common opinions and patterns among respondents, such as the need for water-saving irrigation, interest in automation, and acceptance of smart farming technologies.

No advanced statistical testing was used at this stage because the purpose of the survey was mainly to understand user perception and project relevance, not to prove final system performance.

1.6. Justification of the Methodology

The selected methodology is suitable for this project because the work is currently focused on research, planning, and prototype design. A literature review alone would provide academic background, but it would not show user awareness or practical expectations. Similarly, a survey alone would not be enough to understand the technical background of smart irrigation and robotics.

Therefore, combining literature review with survey analysis provides a stronger foundation for the project. The literature review supports the technical direction, while the survey helps justify the practical need for the proposed Smart Irrigation Rover.

1.7. Obstacles and Solutions

During the research process, one major challenge was finding papers that were directly related to both smart irrigation and agricultural robotics. Many papers focused only on fixed irrigation systems, while others focused only on robotic navigation. To address this issue, the papers were divided into categories such as IoT irrigation, soil moisture monitoring, agricultural robots, obstacle avoidance, and AI-based decision systems.

Another challenge was keeping the project scope realistic. Many advanced systems use expensive technologies such as LiDAR, GPS, drones, or advanced machine learning models. Since this project is intended to be a low-cost academic prototype, the selected design focuses on accessible components such as ESP32, soil moisture sensor, ultrasonic sensor, DHT11 sensor, relay module, motor driver, and water pump.

The survey process also had limitations because the number and background of respondents may not fully represent professional farmers. However, the survey still provides useful feedback for understanding general opinions about smart irrigation and automation.

1.8. Sources Used for Methodology

The methodology was developed using reviewed research papers, survey responses, and project planning documents. The research papers provided the academic and technical foundation, while the Google Form survey provided primary feedback from respondents. These sources helped define the research problem, select the project approach, and identify the design requirements for the proposed Smart Irrigation Rover.

2. Research Questions

RQ1: How can soil moisture sensing help identify whether a specific soil area requires irrigation?

RQ2: How can a mobile rover-based irrigation system improve flexibility compared to fixed smart irrigation systems?

RQ3: How useful is IoT-based monitoring for observing soil moisture, temperature, humidity, and irrigation status remotely?

RQ4: How can obstacle detection support safer movement of an autonomous irrigation rover in a small agricultural environment?

RQ5: How can adaptive spot watering reduce unnecessary water usage compared to traditional manual or uniform irrigation methods?

RQ6: What are the practical challenges of developing a low-cost smart irrigation rover using locally available components?

3. Survey Questionnaire Preparation

The survey questionnaire was prepared to understand people's opinions about irrigation problems, soil monitoring, smart farming technology, and the need for an automated irrigation rover. The questionnaire was created using Google Forms and included demographic, technical, and opinion-based questions. A total of 21 responses were collected. The survey questions were originally prepared with both English and Bangla text in the project proposal form.

3.1. Survey Questions

1. What is your age group?
2. What is your occupation?
3. Are you involved in farming or plant cultivation?
4. Do you think manual irrigation wastes water?
5. Is manual irrigation time-consuming?
6. Do you face difficulties monitoring soil conditions regularly?
7. Do you think overwatering can damage crops or plants?

8. How important is soil moisture monitoring in agriculture?
9. Would an automated irrigation robot help reduce labor?
10. Do you think smart farming technology can improve agriculture?
11. Would you like a system that can automatically detect dry soil and irrigate it?
12. Is obstacle avoidance important for autonomous farming robots?
13. Have you heard about IoT-based smart farming before?
14. Do you believe AI and automation can improve future agriculture?
15. Would you be interested in using low-cost smart farming robots?
16. What are the biggest problems farmers face during irrigation?
17. What features would you expect from a smart farming robot?

4. Prototype 3D Design

The prototype 3D design was prepared to visualize the expected structure of the Smart Irrigation Rover before hardware implementation. The design shows the rover body, water tank, sensor placement, wheels, and irrigation mechanism.

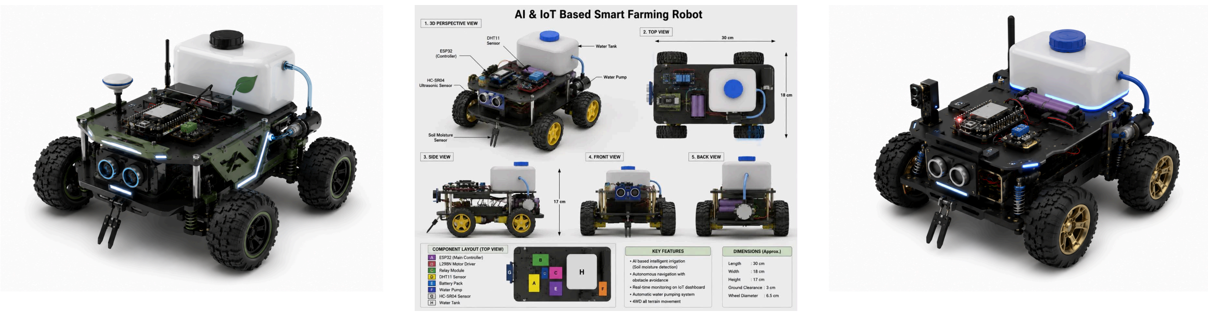

Figure 1: Prototype 3D design views of the Smart Irrigation Rover

5. Component List with Price Estimation

No	Component	Qty	Unit	Total	Purpose
1	4WD Chassis Kit	1	980	980	Robot body
2	ESP32 Board	1	448	448	Main controller
3	L298N Motor Driver	1	199	199	Motor control
4	Ultrasonic Sensor	1	85	85	Obstacle detection
5	DHT11 Sensor	1	148	148	Temp. and humidity
6	Soil Moisture Sensor	1	165	165	Soil detection
7	Relay Module	1	88	88	Pump switching
8	Water Pump	1	65	65	Irrigation
9	Servo Motor	1	190	190	Nozzle movement

10	Buck Converter	1	120	120	Voltage control
11	LCD with I2C	1	240	240	Data display
12	18650 Battery	2	195	390	Power supply
13	Battery Case	1	35	35	Battery holder
14	Battery Charger	1	140	140	Charging
15	Breadboard	1	75	75	Testing
16	Jumper Wires	3	80	240	Wiring
17	LED + Resistor	1	50	50	Indicators
18	Water Pipe	1	80	80	Water delivery
19	Spray Nozzle	1	50	50	Water output
20	Glue Gun + Sticks	1	330	330	Mounting
21	Soldering Items	1	430	430	Wiring work
22	Tape + Heat Shrink	1	110	110	Insulation
23	Screwdriver Set	1	150	150	Assembly
24	Zip Ties	1	50	50	Cable management
25	GPS Module	1	350	350	Future tracking
26	GSM Module	1	450	450	Emergency alert
27	Buzzer + Button	1	100	100	Emergency safety

5.1. Estimated Total

Category	**Cost (৳)**
Current Required Components	5106
Future Optional Components	1200
Approximate Total	6306

6. Classification Tables

6.1. Classification of Reviewed Papers by Research Area

No	**Research Area**	**Level of Relevance**	**Relevance to Project**
1	Smart Irrigation Systems	High	Supports adaptive watering and water-saving irrigation decisions
2	IoT-Based Agriculture	High	Supports real-time monitoring and sensor data communication
3	Agricultural Robotics	High	Supports mobile rover design and autonomous operation
4	Soil Moisture Monitoring	High	Supports dry soil detection and moisture-based irrigation
5	Obstacle Avoidance	Medium	Supports safe rover movement in field conditions
6	AI / ML in Agriculture	Medium	Supports future intelligent decision-making

7	Remote Sensing	Low-Medium	Supports broader precision agriculture understanding

6.2. Classification of Technologies Used

No	Technology	Use in Smart Irrigation Rover
1	IoT	Remote monitoring of soil moisture, temperature, humidity, and pump status
2	Soil Moisture Sensing	Detects dry, moderate, and wet soil conditions
3	Embedded System	ESP32 controls sensors, motor driver, relay, and pump
4	Autonomous Robotics	Allows rover movement through small agricultural areas
5	Obstacle Detection	Ultrasonic sensor helps avoid collisions
6	Water Pump Control	Provides spot watering only when needed
7	AI / Decision Logic	Used for rule-based soil condition classification and future improvement

6.3. Classification of Project Components

No	Component	Category	Function
1	ESP32	Controller	Processes sensor data and controls the system
2	Soil Moisture Sensor	Sensor	Measures soil moisture level
3	DHT11 Sensor	Sensor	Measures temperature and humidity
4	HC-SR04 Sensor	Sensor	Detects obstacles
5	L298N Motor Driver	Actuator Driver	Controls rover motors
6	Relay Module	Switching Module	Switches water pump
7	Water Pump	Actuator	Delivers water to dry soil
8	4WD Chassis	Mechanical Structure	Provides rover movement platform
9	Battery Pack	Power Supply	Provides electrical power
10	IoT Dashboard	Monitoring System	Displays real-time data

6.4. Classification of Data Sources

No	Data Source	Purpose
1	Literature Review	Identifies existing technologies, gaps, and design direction
2	Google Form Survey	Collects user opinions about irrigation and smart farming
3	Component Price List	Estimates project cost and feasibility
4	Prototype Design	Visualizes expected rover structure before implementation

5	Project Proposal	Defines goals, benefits, block diagram, and project scope

7. Data Analysis

The data analysis was completed using the literature review, Google Form survey responses, and component cost estimation. Since the project is still in the design and prototype planning stage, the analysis mainly focused on understanding the problem, user needs, research gaps, and project feasibility.

7.1. Literature Review Analysis

The reviewed papers were grouped into smart irrigation, IoT monitoring, soil moisture sensing, agricultural robotics, obstacle avoidance, and AI-based decision-making. Most papers showed that smart irrigation can reduce water waste, but many systems are fixed and do not move between different soil points. This supports the need for a mobile irrigation rover.

7.2. Survey Analysis

The Google Form survey responses were analyzed using charts and percentages. The survey helped understand people's opinions about manual irrigation, water wastage, smart farming, and automated irrigation systems. The responses supported the idea that a low-cost smart irrigation system can be useful for reducing manual effort and improving water management.

7.3. Feasibility Analysis

The component price estimation showed that the rover can be developed with affordable and locally available components such as ESP32, soil moisture sensor, ultrasonic sensor, motor driver, relay module, water pump, and 4WD chassis. Optional features like GPS and emergency alert can be added later as future improvements.

7.4. Summary

Overall, the analysis shows that the Smart Irrigation Rover is relevant, practical, and feasible. The literature review supports the technical direction, while the survey responses support the real-world need for adaptive spot watering and smart irrigation.

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